SIMATS ENGINEERING
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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Logical Interface Engineering & Protocol Justification

(Linux Netplan YAML, LACP (802.3ad), Layer3+4 Hash Policy)

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Introduction

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What is Netplan?

Linux Netplan:

Netplan is a **network configuration utility** used in Ubuntu Linux.

It uses YAML files to configure:

- IP Address
- Gateway
- DNS
- Ethernet Interfaces
- Bonding
- VLAN
- Bridges

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Netplan YAML Configuration

```
network:
  version: 2

  ethernets:
    eth0:
      dhcp4: no

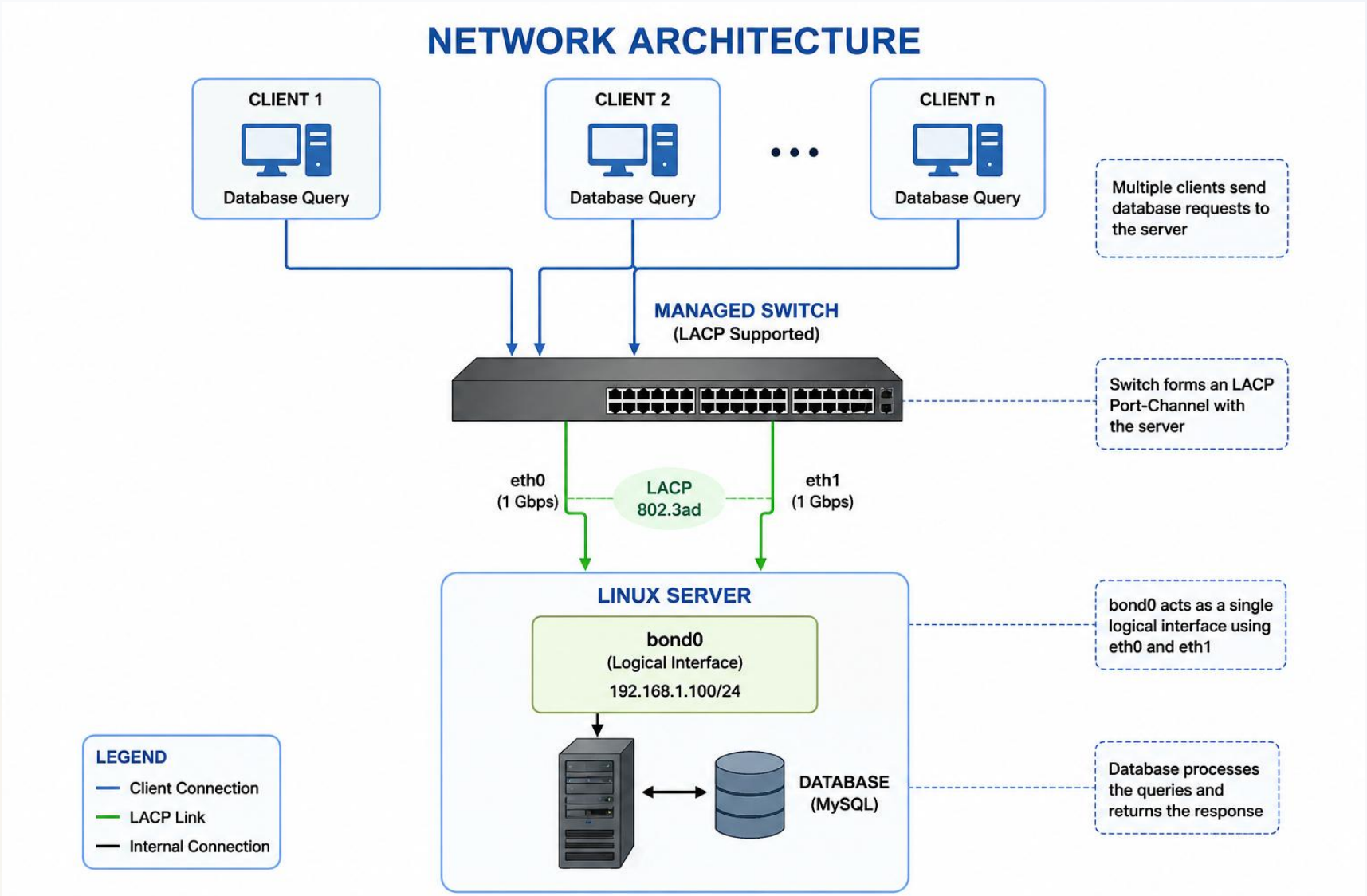
    eth1:
      dhcp4: no

  bonds:
    bond0:
      interfaces:
        - eth0
        - eth1
```

```
      parameters:
        mode: 802.3ad
        lacp-rate: fast
        transmit-hash-policy:
layer3+4

      addresses:
        - 192.168.1.100/24
```

Network Architecture



Why LACP (802.3ad)?

What is LACP?

LACP stands for

Link Aggregation Control Protocol

Why we selected it

- Uses multiple links simultaneously
- Increases bandwidth
- Automatic failover
- Better performance
- Load balancing

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LACP vs Active-Backup

LACP

- Both links active
- Load balancing
- Higher bandwidth
- Automatic failover
- Better performance

Active-Backup

- One link active
- No load balancing
- Normal bandwidth
- Automatic failover
- Lower performance

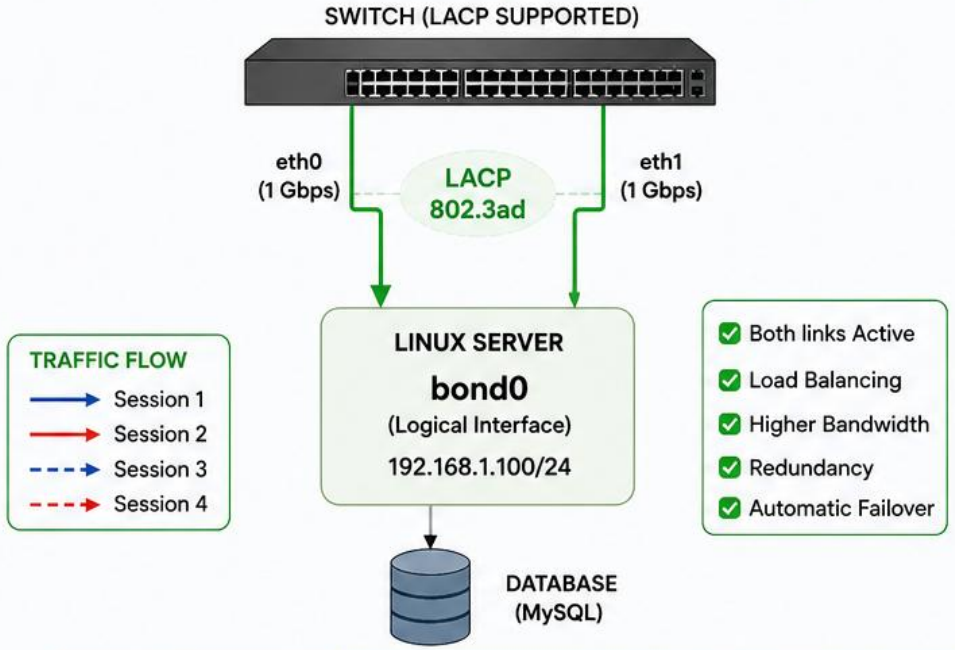
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LACP VS ACTIVE-BACKUP

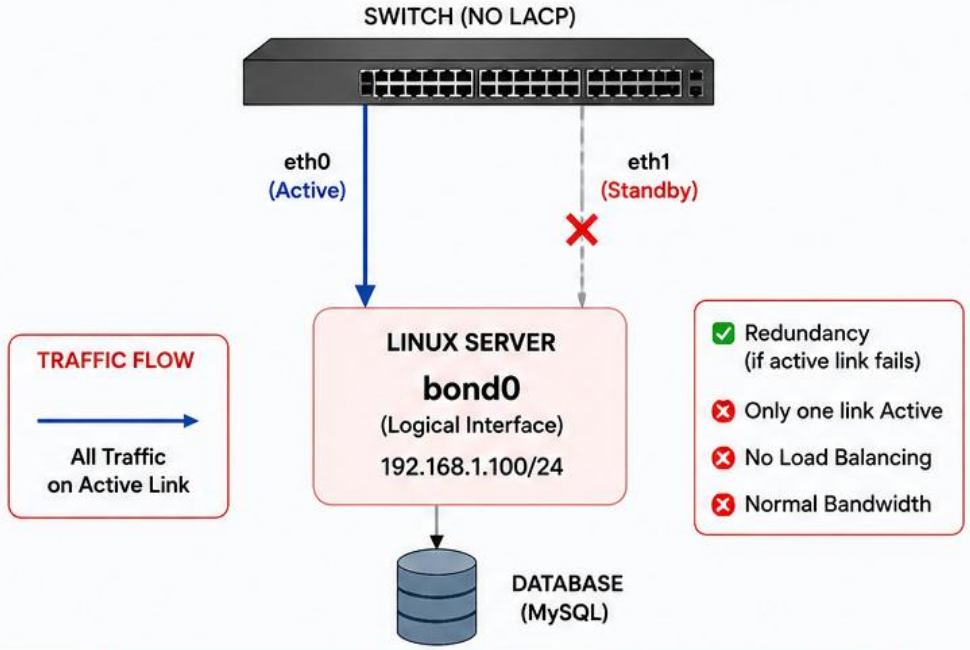
LACP (802.3ad) – Link Aggregation

Both links are active and carry traffic simultaneously.



ACTIVE-BACKUP

Only one link is active. The other link stays in standby.



FEATURE	LACP (802.3ad)	ACTIVE-BACKUP
Links Used	Both links are used simultaneously	Only one link is used at a time
Bandwidth	Increased (Aggregate bandwidth)	Normal (Single link bandwidth)
Load Balancing	Yes (Distributes traffic across links)	No
Redundancy	Yes (If one link fails, other continues)	Yes (Standby link takes over only when failure occurs)
Failover Time	Very Fast (Sub-second with LACP)	Depends on detection (Slightly slower)
Performance	High (Better throughput and efficiency)	Lower (Limited to one link speed)

What is Layer3+4 Hash Policy?

Layer3 means

- Source IP
- Destination IP

Layer4 means

- Source Port
- Destination Port

Linux calculates a hash using these values.

Then chooses

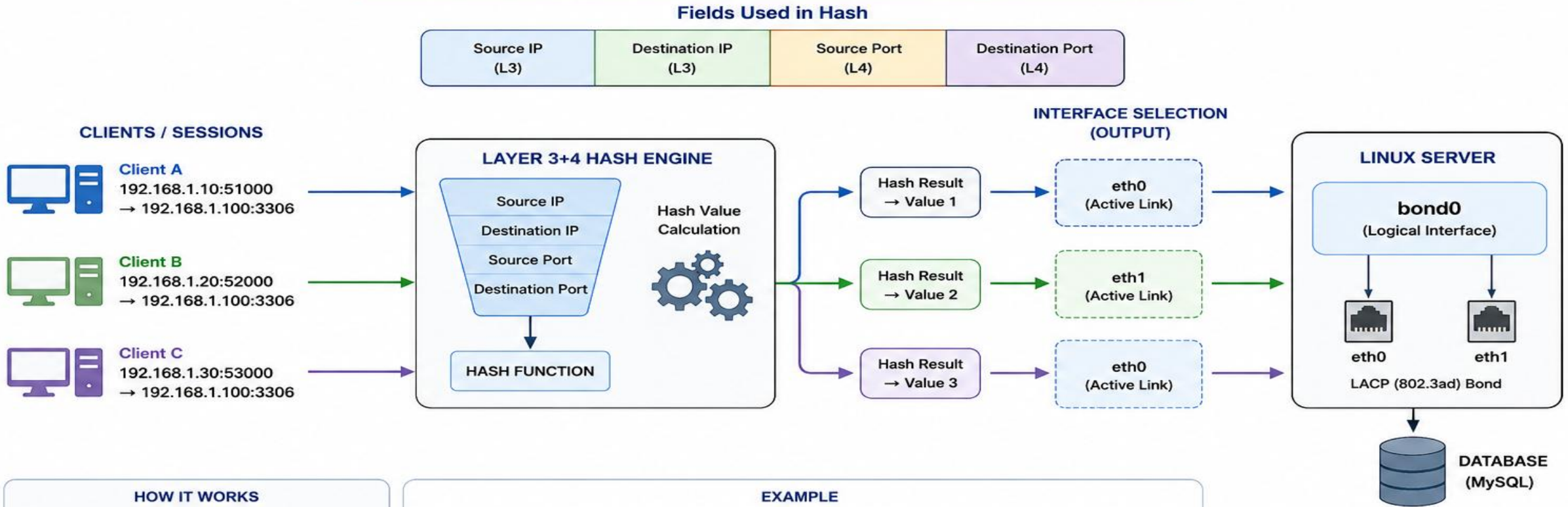
- eth0

  or

- eth1

LAYER 3+4 HASH POLICY – DIAGRAM

Layer 3+4 hash policy uses the following fields to calculate a hash value which decides the outgoing interface (eth0 or eth1).



HOW IT WORKS

1. Each session has unique combination of Source IP, Destination IP, Source Port and Destination Port.
2. These fields are fed into the hash function.
3. A hash value is generated.
4. Based on the hash value, the session is mapped to an interface (eth0 or eth1).
5. All packets of the same session follow the same interface to maintain order.

EXAMPLE

Session	Source IP	Source Port	Destination IP	Destination Port	Hash Value (Example)	Selected Interface
Client A	192.168.1.10	51000	192.168.1.100	3306	25	eth0
Client B	192.168.1.20	52000	192.168.1.100	3306	78	eth1
Client C	192.168.1.30	53000	192.168.1.100	3306	25	eth0

Note: If the same client opens another session with a different source port, the hash value may change and it can be mapped to the other interface.

KEY BENEFITS

- Load balancing across multiple links
- Better bandwidth utilization
- Maintains packet order within a session
- Automatic failover with LACP

In short: Layer 3+4 hash policy distributes different network sessions across available links using IPs and Ports.

Conclusion

- Netplan simplifies Linux network configuration.
- Bonding combines multiple Ethernet interfaces.
- LACP provides higher bandwidth and redundancy.
- Layer3+4 distributes sessions efficiently.
- Database traffic remains stable and balanced

Thank You!